

A working model of behavior leakage: a constant write, gated by context similarity

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1 Setup

Leakage: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appears under other, untrained contexts. We measure it as the trained–base change in the behavior’s expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander’s own response (#502/#489), judge-scored agreement rates for sycophancy (#509), misaligned-and-coherent rates for emergent misalignment (#521), taught-fact emission for fact implants (#500).

We summarize a context c by an activation vector $v_c \in \mathbb{R}^d$ and a behavior by the *behavior state* $v_b(c) = f(c)$, the activations where the behavior is expressed, with f the model’s unknown context-to-behavior map. Each behavior also has a *read-out direction* r_b whose projection tracks expression (the unembedding row for the marker; a probe or contrastive direction otherwise). The constructions of v_c and v_b (layer, position, aggregation) are swept, not fixed; the experiments in §4 search over them.

2 The simplest model

Assumptions.

A1 (behaviors are linear directions).

Adding a single residual-stream direction elicits a behavior [3, 4]; projections onto single directions track traits before generation [5]; emergent misalignment is predicted, controlled, and ablated through one direction [6, 7].

A2 (edits are read/write pairs).

Weight updates read input directions and write output directions, with the rank-one outer product as the atomic unit [1]; low-rank adapters make this explicit [16, 2]. Narrow fine-tunes act as a near-constant steering write [8, 9], and the parameterization is irrelevant: at matched implant strength, LoRA and full fine-tuning leak identically (gap +0.00 nat, #514).

A3 (linearity, the extra assumption).

At one layer, $v_b \approx Mv_c$, and the edit acts at that layer.

A4 (minimal edit).

Training implements the smallest weight change that fits the new pair, the rank-one key→value association of linear associative memories and ROME [10, 11, 12].

From assumptions to formula. Fine-tune context c' from behavior state $v_{b'} = f(c')$ to an achieved state $v_{b''}$ (under early stopping the achieved state undershoots the target, so the

write is measured, not assumed). A4 with A3 gives the edit: minimizing $\|\Delta M\|_F$ subject to $(M+\Delta M)v_{c'} = v_{b'}$ yields

$$M' = M + \frac{(v_{b''} - v_{b'})v_{c'}^\top}{\|v_{c'}\|^2} \implies \Delta v_b(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what} \text{ leaks}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{gate: } \textit{who} \text{ leaks}}. \quad (1)$$

Each assumption does one job: A4 makes the edit rank-one (one write direction, one key), A3 makes the gate the activation inner product, A2 makes the same structure visible in the weights ($\Delta W \approx \text{write} \otimes \text{key}$), and A1 makes the write measurable, since the observed change is one slope per behavior times the gate: $\Delta z_b(c) = \langle r_b, v_{b''} - v_{b'} \rangle g(c)$. The softmax compresses log-probabilities near saturation and can reorder contexts there, which is why we store logits alongside log-probabilities.

3 Predictions

P0 (weights decompose).

The top singular vectors of the trained ΔW are the key and the write. Under contrastive training the key moves from raw $v_{c'}$ toward a source-minus-negatives contrast direction as dose grows.

P1 (one write direction).

Per-bystander activation shifts at the read-out all point along $(v_{b''} - v_{b'})$; an SVD of the shift matrix is near rank-one with a top component matching the behavior delta.

P2 (the gate is base-model context similarity).

Some fixed kernel on base-model context summaries sets who leaks, and it transfers across held-out panels. On the linear rung it is the dot product, hence asymmetric: $c' \rightarrow c$ scales as $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$.

P3 (the behavior's prior matters, through three channels).

Source side: the write scales with the residual $\|v_{b''} - v_{b'}\|$, so a source that already performs the behavior produces little leakage despite high similarity. Bystander side: the prior enters through read-out compression (measurement) and possibly through the gate itself (mechanism); the two separate in logit space, but only with the exact gate rather than a proxy.

P4 (additivity).

Two implants trained jointly leak like the sum of their singletons, with deviations set by the sources' mutual similarity (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).

The gate predicts best at the layers where the behavior is computed; scoring this needs an independent localization (patching or probing), since scoring by predictor quality is circular.

P6 (cross-behavior transfer).

The same write moves other behaviors in proportion to its projection onto their read-out directions, extending the rule from context generalization to behavior generalization.

P7 (ICL proxy).

The base model's in-context behavior on the training mix forward-predicts the post-training leakage map (see §5 for why).

No single miss refutes the model, since each prediction has a relaxation (§5); the conjunction does. With the gate computed exactly and the write measured on a rig-clean implant, a failure of write $\times$ gate to rank leakage on a held-out panel leaves no version standing.

4 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]’s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction (cos 0.96–0.98); the marker implant does not, and its trained context is <i>least</i> aligned. Benign SFT also gives one direction (#552, running); positive-only retrain tests the rig confound (#561, running).
P2	Correlate base-model similarity with measured leakage across source $\times$ bystander panels (#502, #489, #509)	Similarity ranks marker leakage ($ \rho \approx 0.76$ –0.79, layers 19–24) and extends to ICL/multi-turn contexts, but only ordinally (out-of-fold $R^2 \approx 0$, single-seed); structure does not transfer across panels (#553); the fact arm reversed sign ($ \rho = 0.49$) while the base prior predicted correctly (+0.27, #500), an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander’s own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); the measurement-vs-mechanism split awaits the exact gate.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker predicts best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as a leakage predictor	Queued; the similarity gate already extends to ICL source contexts (#489).

The open headline experiment is a dose titration: implant at a non-saturated anchor (#448), measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432 $\rightarrow$ #456), at several doses across the 10–25-step cliff (#139), with at least 3 seeds. At low dose the shift matrix should be near rank-one with similarity-graded magnitude; across the cliff the direction should rotate and the prediction collapse. The P1 split already complicates the low-dose half, so the titration doubles as its diagnosis.

5 Relaxing the assumptions

Drop A3 (linearity). For a small update, the change at any input is first-order: $\Delta f(c) \approx K(c, c') \times \text{residual}$, with K the empirical tangent kernel [14], a regime measured to hold for language-model fine-tuning in several settings [15]. The formula survives with $g(c) = K(c, c')/K(c', c')$, where K is the cross-kernel between the read-out gradient at c and the training-loss gradient at c' (symmetric only for read-out-aligned losses such as marker-only). The dot product of equation (1) is the linear-kernel case, and the practical ladder runs cosine $\rightarrow$ distributional comparisons (Gaussian-KL, MMD) $\rightarrow$ the exact cross-kernel via backprop. Gains over cosine have so far been marginal (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy).

Relax A4 (rank-one). The write becomes a low-rank subspace, $\Delta v_b(c) = U \text{diag}(s) V^\top v_c$; P1 weakens to “shifts span r dimensions,” and the SVD already in use measures r .

Relax the gate’s form. The gate need not be a dot product: a constant vector can carry a fully conditional behavior by shifting queries into pre-existing attention circuitry that fetches the behavior from the context’s own values [8]. The write is then a router, not a copy of the behavior direction; the cleanest evidence is a rank-1 EM adapter whose B vector has cosine 0.04 to the misalignment direction at its own layer while 98% of its effect ablates along that direction downstream [7]. Under routing, the gate is attention-shaped (near-binary for discrete triggers) and becomes estimable from samples rather than computable from cosines, and P1 can fail by mechanism when fetched content differs across contexts.

Multiple training pairs. The edit solves a kernel system over all pairs; contrastive negatives enter as off-diagonal terms that subtract the common mode (persona contexts sit near cosine 0.92, which is why positive-only implants leak near-uniformly while contrastive ones localize: $\sim 99.6\%$ source vs $\sim 11.7\%$ bystander, #247/#329). Early stopping spectrally filters the solution toward raw similarity, so the time-indexed predictor interpolates between raw and Gram-corrected similarity; neither endpoint has been run.

Extend to in-context examples. A transformer block provably converts context into a transient rank-1 weight patch [17, 18], so an ICL prefix performs the same write $\times$ gate operation for the duration of the prompt, and the gate already ranks leakage from ICL sources (#489). Full ICL=fine-tuning equivalence remains open for pretrained models [19, 20, 21], but in-context behavior carries predictive signal about fine-tuning outcomes: ICL recovers $> 85\%$ of fine-tuning’s corrections in-distribution [19], in-context inferences anticipate and repair fine-tuning’s generalization failures [22], and demonstration effects compress to a single latent direction [23]. Hence P7.

What cannot be relaxed. Every version assumes the update is small in weight/feature space, though not in behavior: read-outs amplify, so a lazy edit can flip emission discretely or induce EM through nine rank-1 adapters [7], and tripling implant depth left the edit geometry unchanged (#538). If training moves the features themselves (the rich regime), no base-model predictor can work, and the historical EM predictor failure looks like that regime ($\rho = 0.09$ prose-only #458; assistant-axis rotation $38\text{--}53^\circ$, discrimination collapse, #125). The current tension: the marker arm passes the gate test but fails the one-direction test, while EM does the reverse; #552 and #561 decide whether the rig or the regime explains it. The durable question under all versions is the same: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

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